

K-Nearest Neighbors (KNN)

on the Diabetes Dataset

Topic: Machine Learning – K-Nearest Neighbors

CS ELEC 01 : Computational Science

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Part 1: Data Understanding

1.1 Feature Descriptions

The dataset contains 768 patient records with 9 columns. Each feature is described below in plain medical and data-science terms.

Feature	Type	Description
Pregnancies	Integer	Number of times the patient has been pregnant. Higher parity is associated with gestational diabetes risk.
Glucose	Continuous	Plasma glucose concentration (mg/dL) measured 2 hours into an oral glucose tolerance test. Primary biomarker for diabetes diagnosis.
BloodPressure	Continuous	Diastolic blood pressure in mm Hg. Hypertension is closely linked to insulin resistance.
SkinThickness	Continuous	Triceps skin-fold thickness in mm. Used as a proxy for subcutaneous body-fat percentage.
Insulin	Continuous	2-hour serum insulin level (μU/mL). Measures the pancreatic response to glucose load.
BMI	Continuous	Body Mass Index = weight (kg) ÷ height (m) ² . Obesity (BMI ≥ 30) is a primary risk factor for Type-2 diabetes.
DiabetesPedigreeFunction	Continuous	A genetic risk score synthesizing family history of diabetes. Higher values indicate stronger hereditary predisposition.
Age	Integer	Age of the patient in years. Diabetes prevalence increases with age.
Outcome	Binary	Target variable: 0 = Non-diabetic, 1 = Diabetic.

1.2 Feature Importance Analysis

Features most likely important for prediction:

- Glucose — the single strongest predictor; elevated plasma glucose directly indicates impaired glucose metabolism.
- BMI — obesity strongly predicts insulin resistance and Type-2 diabetes.
- Age — older patients have higher cumulative risk of metabolic syndrome.
- DiabetesPedigreeFunction — captures genetic risk that other clinical measurements may miss.
- Insulin — directly quantifies the pancreas's ability to regulate blood sugar.

1.3 Missing / Problematic Values

Features likely containing missing data (encoded as zero):

- Glucose — a plasma glucose of 0 is physiologically impossible.
- BloodPressure — a diastolic BP of 0 implies cardiac arrest; must be missing.
- SkinThickness — 0 mm is anatomically impossible.
- Insulin — 0 μU/mL serum insulin in a living patient is unrealistic; 374 records (48.7%) are zero.

- BMI — a BMI of 0 is impossible for a living human.

Pregnancies = 0 is valid (nulliparous patient). Outcome = 0 is the Non-diabetic class label, not missing.

Part 2: Data Preprocessing

2.1 Zero-Value Scan

Total records: 768. The table below counts zeros per feature (zeros in medical features = missing values).

Feature	Zero Count	Percentage	Biologically Valid?
Pregnancies	111	14.5%	✓ Valid
Glucose	5	0.7%	X Missing
BloodPressure	35	4.6%	X Missing
SkinThickness	227	29.6%	X Missing
Insulin	374	48.7%	X Missing
BMI	11	1.4%	X Missing
DiabetesPedigreeFunction	0	0.0%	✓ Valid
Age	0	0.0%	✓ Valid

2.2 Summary Statistics — Before Preprocessing

Raw dataset statistics. Notice zeros pulling means and minimums down for Glucose, BloodPressure, SkinThickness, Insulin, and BMI.

Feature	Min	Max	Mean	Std Dev
Pregnancies	0.00	17.00	3.8451	3.3674
Glucose	0.00	199.00	120.8945	31.9518
BloodPressure	0.00	122.00	69.1055	19.3432
SkinThickness	0.00	99.00	20.5365	15.9418
Insulin	0.00	846.00	79.7995	115.1689
BMI	0.00	67.10	31.9926	7.8790
DiabetesPedigreeFunction	0.08	2.42	0.4719	0.3311
Age	21.00	81.00	33.2409	11.7526

2.3 Missing Data Handling — Median Imputation

Strategy chosen: Median Imputation. The dataset was first shuffled and split into an 80% training set and a 20% test set. Medians were then computed exclusively from the non-zero values in the training set. These training-set medians were applied to replace zeros in both splits. This order is critical: computing medians on the full dataset before splitting constitutes data leakage, as test-set values

would influence the imputation applied to test data. Median is preferred over mean because it is robust to outliers (e.g. extreme insulin spikes).

Imputation Formula

For each row r and imputed feature f :

If $r[f] == 0 \rightarrow r[f] = \text{median}(\{x \in \text{feature_f} \mid x \neq 0\})$

Computed Medians (Training Set Only)

Feature	Non-zero Median	Zeros Replaced
Glucose	117.00	5
BloodPressure	72.00	35
SkinThickness	29.00	227
Insulin	125.00	374
BMI	32.30	11

2.4 Summary Statistics — After Preprocessing

After imputation, all minimums are positive. Means and standard deviations shift toward realistic physiological ranges.

Feature	Min	Max	Mean	Std Dev
Pregnancies	0.00	17.00	3.8451	3.3674
Glucose	44.00	199.00	121.6562	30.4185
BloodPressure	24.00	122.00	72.3867	12.0888
SkinThickness	7.00	99.00	29.1081	8.7855
Insulin	14.00	846.00	140.6719	86.3268
BMI	18.20	67.10	32.4552	6.8707
DiabetesPedigreeFunction	0.08	2.42	0.4719	0.3311
Age	21.00	81.00	33.2409	11.7526

2.5 Feature Scaling — Min-Max Normalisation

KNN computes Euclidean distance across all features simultaneously. Without scaling, Insulin (0–846) would dominate Pregnancies (0–17) by a factor of $\sim 50\times$, making the latter nearly irrelevant. Min-Max normalisation maps every feature to $[0, 1]$. Critically, the minimum and maximum bounds are derived solely from the training set after imputation, and then applied to both the training and test sets. Using the full dataset to compute scaling bounds would leak test-set range information into the normalisation step.

Formula

$$x_norm = (x - \min(f)) / (\max(f) - \min(f))$$

Min and Max Values Used

Feature	Min (training set)	Max (training set)
Pregnancies	0.0000	17.0000
Glucose	44.0000	199.0000
BloodPressure	24.0000	122.0000
SkinThickness	7.0000	99.0000
Insulin	14.0000	846.0000
BMI	18.2000	67.1000
DiabetesPedigreeFunction	0.0780	2.4200
Age	21.0000	81.0000

Part 3: KNN Implementation

3.1 Dataset Split

Split	Samples	Percentage
Total	768	100%
Training set	614	80%
Test set	154	20%

A fixed random seed (seed = 42) is used to ensure reproducibility across runs.

3.2 Euclidean Distance Formula

$$d(p, q) = \sqrt{\sum_i (p_i - q_i)^2}$$

where p is the test instance and q is a training instance, both in normalised feature space.

3.3 Manual Distance Computation — 10 Training Samples

Test Instance #1 (normalised features):

Feature	Normalised Value
Pregnancies	0.058824
Glucose	0.290323

Feature	Normalised Value
BloodPressure	0.428571
SkinThickness	0.173913
Insulin	0.096154
BMI	0.202454
DiabetesPedigreeFunction	0.038002
Age	0.000000

Actual Outcome: 0 (Non-diabetic)

Step-by-Step Distance Computations

For each of the 10 nearest neighbours, the squared difference per feature is computed, summed, then square-rooted:

Neighbour #1 — Class: 0 (Non-diabetic) $d = 0.052052$

Feature	Test (p)	Neighbour (q)	(p - q)	(p - q) ²
Pregnancies	0.058824	0.058824	0.000000	0.00000000
Glucose	0.290323	0.303226	-0.012903	0.00016649
BloodPressure	0.428571	0.408163	0.020408	0.00041649
SkinThickness	0.173913	0.184783	-0.010870	0.00011816
Insulin	0.096154	0.133413	-0.037259	0.00138823
BMI	0.202454	0.224949	-0.022495	0.00050603
DiabetesPedigreeFunction	0.038002	0.048676	-0.010674	0.00011393
Age	0.000000	0.000000	0.000000	0.00000000

Sum of squares: $0.000000 + 0.000166 + 0.000416 + 0.000118 + 0.001388 + 0.000506 + 0.000114 + 0.000000$

$= 0.00270932$

$d = \sqrt{(0.00270932)} = 0.052051$

Neighbour #2 — Class: 0 (Non-diabetic) $d = 0.112164$

Feature	Test (p)	Neighbour (q)	(p - q)	(p - q) ²
Pregnancies	0.058824	0.117647	-0.058823	0.00346015
Glucose	0.290323	0.296774	-0.006451	0.00004162
BloodPressure	0.428571	0.469388	-0.040817	0.00166603
SkinThickness	0.173913	0.108696	0.065217	0.00425326
Insulin	0.096154	0.133413	-0.037259	0.00138823
BMI	0.202454	0.186094	0.016360	0.00026765
DiabetesPedigreeFunction	0.038002	0.002989	0.035013	0.00122591

Feature	Test (p)	Neighbour (q)	(p - q)	(p - q) ²
Age	0.000000	0.016667	-0.016667	0.00027779

Sum of squares: 0.003460 + 0.000042 + 0.001666 + 0.004253 + 0.001388 + 0.000268 + 0.001226 + 0.000278

= 0.01258063

d = $\sqrt{(0.01258063)} = 0.112163$

Neighbour #3 — Class: 0 (Non-diabetic) d = 0.125705

Feature	Test (p)	Neighbour (q)	(p - q)	(p - q) ²
Pregnancies	0.058824	0.117647	-0.058823	0.00346015
Glucose	0.290323	0.283871	0.006452	0.00004163
BloodPressure	0.428571	0.510204	-0.081633	0.00666395
SkinThickness	0.173913	0.130435	0.043478	0.00189034
Insulin	0.096154	0.046875	0.049279	0.00242842
BMI	0.202454	0.220859	-0.018405	0.00033874
DiabetesPedigreeFunction	0.038002	0.064475	-0.026473	0.00070082
Age	0.000000	0.016667	-0.016667	0.00027779

Sum of squares: 0.003460 + 0.000042 + 0.006664 + 0.001890 + 0.002428 + 0.000339 + 0.000701 + 0.000278

= 0.01580183

d = $\sqrt{(0.01580183)} = 0.125705$

Neighbour #4 — Class: 0 (Non-diabetic) d = 0.133099

Feature	Test (p)	Neighbour (q)	(p - q)	(p - q) ²
Pregnancies	0.058824	0.117647	-0.058823	0.00346015
Glucose	0.290323	0.309677	-0.019354	0.00037458
BloodPressure	0.428571	0.387755	0.040816	0.00166595
SkinThickness	0.173913	0.228261	-0.054348	0.00295371
Insulin	0.096154	0.133413	-0.037259	0.00138823
BMI	0.202454	0.274029	-0.071575	0.00512298
DiabetesPedigreeFunction	0.038002	0.022203	0.015799	0.00024961
Age	0.000000	0.050000	-0.050000	0.00250000

Sum of squares: 0.003460 + 0.000375 + 0.001666 + 0.002954 + 0.001388 + 0.005123 + 0.000250 + 0.002500

= 0.01771520

d = $\sqrt{(0.01771520)} = 0.133098$

Neighbour #5 — Class: 0 (Non-diabetic) d = 0.134194

Feature	Test (p)	Neighbour (q)	(p - q)	(p - q) ²
Pregnancies	0.058824	0.058824	0.000000	0.00000000
Glucose	0.290323	0.335484	-0.045161	0.00203952
BloodPressure	0.428571	0.408163	0.020408	0.00041649
SkinThickness	0.173913	0.217391	-0.043478	0.00189034
Insulin	0.096154	0.087740	0.008414	0.00007080
BMI	0.202454	0.306748	-0.104294	0.01087724
DiabetesPedigreeFunction	0.038002	0.090094	-0.052092	0.00271358
Age	0.000000	0.000000	0.000000	0.00000000

Sum of squares: 0.000000 + 0.002040 + 0.000416 + 0.001890 + 0.000071 + 0.010877 + 0.002714 + 0.000000

= 0.01800795

d = $\sqrt{(0.01800795)} = 0.134194$

Neighbour #6 — Class: 0 (Non-diabetic) d = 0.136376

Feature	Test (p)	Neighbour (q)	(p - q)	(p - q) ²
Pregnancies	0.058824	0.058824	0.000000	0.00000000
Glucose	0.290323	0.316129	-0.025806	0.00066595
BloodPressure	0.428571	0.469388	-0.040817	0.00166603
SkinThickness	0.173913	0.260870	-0.086957	0.00756152
Insulin	0.096154	0.133413	-0.037259	0.00138823
BMI	0.202454	0.249489	-0.047035	0.00221229
DiabetesPedigreeFunction	0.038002	0.101196	-0.063194	0.00399348
Age	0.000000	0.033333	-0.033333	0.00111109

Sum of squares: 0.000000 + 0.000666 + 0.001666 + 0.007562 + 0.001388 + 0.002212 + 0.003993 + 0.001111

= 0.01859859

d = $\sqrt{(0.01859859)} = 0.136377$

Neighbour #7 — Class: 0 (Non-diabetic) d = 0.136560

Feature	Test (p)	Neighbour (q)	(p - q)	(p - q) ²
Pregnancies	0.058824	0.117647	-0.058823	0.00346015
Glucose	0.290323	0.238710	0.051613	0.00266390
BloodPressure	0.428571	0.367347	0.061224	0.00374838
SkinThickness	0.173913	0.163043	0.010870	0.00011816
Insulin	0.096154	0.133413	-0.037259	0.00138823
BMI	0.202454	0.194274	0.008180	0.00006691
DiabetesPedigreeFunction	0.038002	0.090521	-0.052519	0.00275825

Feature	Test (p)	Neighbour (q)	(p - q)	(p - q) ²
Age	0.000000	0.066667	-0.066667	0.00444449

Sum of squares: 0.003460 + 0.002664 + 0.003748 + 0.000118 + 0.001388 + 0.000067 + 0.002758 + 0.004444

= 0.01864846

d = $\sqrt{(0.01864846)} = 0.136559$

Neighbour #8 — Class: 0 (Non-diabetic) d = 0.138940

Feature	Test (p)	Neighbour (q)	(p - q)	(p - q) ²
Pregnancies	0.058824	0.058824	0.000000	0.00000000
Glucose	0.290323	0.329032	-0.038709	0.00149839
BloodPressure	0.428571	0.367347	0.061224	0.00374838
SkinThickness	0.173913	0.119565	0.054348	0.00295371
Insulin	0.096154	0.052885	0.043269	0.00187221
BMI	0.202454	0.116564	0.085890	0.00737709
DiabetesPedigreeFunction	0.038002	0.077711	-0.039709	0.00157680
Age	0.000000	0.016667	-0.016667	0.00027779

Sum of squares: 0.000000 + 0.001498 + 0.003748 + 0.002954 + 0.001872 + 0.007377 + 0.001577 + 0.000278

= 0.01930436

d = $\sqrt{(0.01930436)} = 0.138940$

Neighbour #9 — Class: 0 (Non-diabetic) d = 0.140565

Feature	Test (p)	Neighbour (q)	(p - q)	(p - q) ²
Pregnancies	0.058824	0.058824	0.000000	0.00000000
Glucose	0.290323	0.283871	0.006452	0.00004163
BloodPressure	0.428571	0.387755	0.040816	0.00166595
SkinThickness	0.173913	0.184783	-0.010870	0.00011816
Insulin	0.096154	0.036058	0.060096	0.00361153
BMI	0.202454	0.239264	-0.036810	0.00135498
DiabetesPedigreeFunction	0.038002	0.146883	-0.108881	0.01185507
Age	0.000000	0.033333	-0.033333	0.00111109

Sum of squares: 0.000000 + 0.000042 + 0.001666 + 0.000118 + 0.003612 + 0.001355 + 0.011855 + 0.001111

= 0.01975840

d = $\sqrt{(0.01975840)} = 0.140565$

Neighbour #10 — Class: 0 (Non-diabetic) d = 0.142106

Feature	Test (p)	Neighbour (q)	(p - q)	(p - q) ²
Pregnancies	0.058824	0.000000	0.058824	0.00346026
Glucose	0.290323	0.367742	-0.077419	0.00599370
BloodPressure	0.428571	0.418367	0.010204	0.00010412
SkinThickness	0.173913	0.228261	-0.054348	0.00295371
Insulin	0.096154	0.133413	-0.037259	0.00138823
BMI	0.202454	0.130879	0.071575	0.00512298
DiabetesPedigreeFunction	0.038002	0.067891	-0.029889	0.00089335
Age	0.000000	0.016667	-0.016667	0.00027779

Sum of squares: 0.003460 + 0.005994 + 0.000104 + 0.002954 + 0.001388 + 0.005123 + 0.000893 + 0.000278

= 0.02019415

d = $\sqrt{(0.02019415)} = 0.142106$

3.4 KNN Results for K = 3, 5, 7

Medians and scaling bounds now derived from training set only (split-first pipeline).

K	TP	TN	FP	FN	Accuracy	Precision	Recall
3	30	80	22	22	71.43%	57.69%	57.69%
5 ★	28	83	19	24	72.08%	59.57%	53.85%
7	26	84	18	26	71.43%	59.09%	50.00%

★ Best K = 5 with 72.08% accuracy. K = 3 and K = 7 both achieve 71.43%. K = 5 is selected as the optimal value.

Part 4: Model Evaluation

4.1 Confusion Matrices

K = 3 — Accuracy: 71.43%

	Predicted: 0 (Non-diabetic)	Predicted: 1 (Diabetic)
Actual: 0 (Non-diabetic)	TN = 80	FP = 22
Actual: 1 (Diabetic)	FN = 22	TP = 30

K = 5 — Accuracy: 72.08% ← Best K

	Predicted: 0 (Non-diabetic)	Predicted: 1 (Diabetic)
Actual: 0 (Non-diabetic)	TN = 83	FP = 19
Actual: 1 (Diabetic)	FN = 24	TP = 28

K = 7 — Accuracy: 71.43%

	Predicted: 0 (Non-diabetic)	Predicted: 1 (Diabetic)
Actual: 0 (Non-diabetic)	TN = 84	FP = 18
Actual: 1 (Diabetic)	FN = 26	TP = 26

4.2 Evaluation Analysis

Q1 — Which K performed best?

K = 5 achieved the highest accuracy of 72.08% on the test set. While K = 7 produced the fewest false positives (FP = 18) and highest TN (84), K = 5 strikes the better overall balance — it outperforms K = 3 on specificity and matches K = 3's lead on accuracy. K = 5 is therefore selected as the optimal value.

Q2 — Why does performance change with different K values?

KNN assigns the predicted class by majority vote among the K nearest neighbours. When K is small, the decision depends on very few neighbours, making it sensitive to local noise and outliers (high variance). When K is large, the vote is spread over many neighbours, smoothing out irregularities but potentially ignoring meaningful local patterns (high bias). The ideal K balances this trade-off for the dataset.

Q3 — What happens when K is too small or too large?

- **Too small (e.g. $K = 1$):** The model memorises training noise — extreme overfitting. A single mislabelled neighbour determines the outcome. Training accuracy may be perfect but test accuracy drops sharply.
- **Too large (e.g. $K = n$):** Every query returns the majority class of the entire training set regardless of local structure — underfitting. The model effectively becomes a constant classifier.

Part 5: Analysis and Reflection

5.1 Strengths of KNN

- Simple and intuitive: the decision logic ("look at the K most similar examples and vote") is easily explained to non-technical stakeholders.
- Non-parametric: KNN makes no assumptions about the underlying data distribution. It can capture arbitrary, non-linear decision boundaries.
- No training phase (lazy learning): the model stores training data verbatim. Adding new training examples requires no retraining — just appending to the dataset.
- Naturally handles multi-class classification without modification.
- Works well on small to medium datasets where prediction latency is not critical.

5.2 Limitations of KNN

- Computational cost at prediction time: every prediction requires computing distances to all n training samples, making it $O(n \cdot d)$ per query — slow for large datasets.
- High memory usage: the entire training dataset must be held in memory.
- Sensitivity to feature scaling: without normalisation, high-range features (e.g. Insulin: 14–846) completely overshadow low-range ones (Pregnancies: 0–17). Min-Max normalisation was essential in this experiment.
- Curse of dimensionality: as the number of features grows, distances become less meaningful — points become approximately equidistant, and nearest-neighbour search loses discriminative power.
- Sensitivity to irrelevant and redundant features: including noisy features increases distances uniformly and degrades classification.
- Class imbalance: if one class dominates the training set (65% Non-diabetic here), majority voting naturally biases toward that class, causing poor recall for the minority class.

5.3 When KNN Is Appropriate

- Small to medium datasets where $O(n)$ prediction cost is acceptable.
- Problems where the decision boundary is complex and non-linear.
- When interpretability of individual predictions matters (each prediction traceable to concrete similar examples).
- Recommendation systems, anomaly detection, and image retrieval tasks.

5.4 When KNN Is Not Appropriate

- Large-scale, real-time systems where latency is critical (e.g. fraud detection at scale).
- High-dimensional data (images, NLP embeddings) where the curse of dimensionality applies.
- Severely imbalanced datasets without additional handling (oversampling, weighted voting).

- Streaming data environments where the training set changes continuously.

5.5 Observations from This Experiment

Median imputation had a notable positive effect. Before imputation, 374 of 768 Insulin values (48.7%) were zero — a biologically impossible value that introduced systematic distortion. These were replaced using the median of non-zero Insulin values in the training set, computed after the 80/20 split. Critically, the split was performed before any imputation or scaling, ensuring that training-set statistics were never contaminated by test-set values. This corrected approach produced more realistic and generalizable results compared to an earlier version that imputed and scaled before splitting — which artificially inflated accuracy by giving the model a statistical “cheat sheet” about the test data.

Min-Max normalisation was essential. Without it, the Insulin feature (range: 14–846) would contribute ~50× more to the Euclidean distance than Pregnancies (range: 0–17). Normalisation ensured all features contributed proportionally, which is a fundamental requirement of any distance-based algorithm.

K = 3 and K = 7 tied on overall accuracy (72.08%) on this test split. K = 3 captured slightly more true diabetics (higher recall: 61.5% vs 53.8%), which is clinically more desirable — missing a diabetic patient (false negative) is riskier than a false alarm. K = 7 achieved marginally higher precision (59.6% vs 58.2%), meaning fewer healthy patients were misclassified as diabetic.

Class imbalance (65% non-diabetic) remains the primary challenge. KNN's majority vote naturally gravitates toward the dominant class. Future work should explore class-weighted voting, SMOTE oversampling, or threshold adjustment to improve recall for diabetic patients (Class 1).

Bonus: Logistic Regression Comparison

As a bonus, a Logistic Regression classifier was implemented from scratch using gradient descent (no external ML libraries) and trained on the same normalized 80/20 split used for KNN. This allows a direct, fair comparison between the two approaches.

Methodology

Logistic Regression models the probability of a binary outcome using the sigmoid function:

$$P(y = 1 | x) = 1 / (1 + e^{-(w \cdot x + b)})$$

Parameters (weights and bias) were optimized using batch gradient descent with a learning rate of 0.1 over 1000 epochs. The decision threshold is 0.5.

Learned Feature Weights

Higher absolute weight = greater influence on the prediction. Positive weights push toward Diabetic (1); negative weights push toward Non-diabetic (0).

Feature	Weight
Pregnancies	0.856432
Glucose	2.094395
BloodPressure	-0.357028
SkinThickness	0.179147
Insulin	0.301519
BMI	1.092437
DiabetesPedigreeFunction	0.576155
Age	0.730485
Bias	-2.383663

Glucose (2.094) and BMI (1.092) are the strongest positive predictors, consistent with the feature importance analysis in Part 1. BloodPressure carries a slight negative weight, suggesting it is a weak or confounding predictor in this dataset.

Head-to-Head Comparison: KNN (K = 5) vs Logistic Regression

Metric	KNN (K = 5)	Logistic Regression	Better
Accuracy	72.08%	76.62%	Logistic Reg.
Precision	59.57%	76.67%	Logistic Reg.
Recall	53.85%	44.23%	KNN ★
F1 Score	56.57%	56.10%	KNN (narrow)
TP	28	23	KNN
TN	83	95	Logistic Reg.
FP	19	7	Logistic Reg.
FN	24	29	KNN

Analysis

Logistic Regression achieves higher overall accuracy (76.62% vs 72.08%) and substantially better precision (76.67% vs 59.57%), meaning it produces far fewer false alarms — only 7 healthy patients misclassified as diabetic, compared to 19 for KNN.

However, KNN outperforms Logistic Regression on recall (53.85% vs 44.23%). In a medical context, recall is the more critical metric: a false negative means a diabetic patient is sent away undiagnosed. KNN catches more true diabetics (28 vs 23 TP), making it the more conservative, clinically safer choice despite its lower accuracy.

The F1 scores are nearly identical (56.57% vs 56.10%), confirming that neither model decisively dominates when precision and recall are considered together. The choice between them depends on clinical priority: if minimizing false positives matters (e.g. avoiding unnecessary follow-up tests), Logistic Regression is preferable. If minimizing missed diagnoses matters, KNN is preferable.

This difference in behavior reflects a fundamental algorithmic distinction: Logistic Regression draws a single global decision boundary across the feature space, which works well when classes are roughly linearly separable. KNN makes purely local decisions based on proximity, which can be more flexible but also more sensitive to class imbalance — the 65/35 non-diabetic/diabetic split in this dataset may bias KNN's majority vote toward the dominant class, suppressing recall.

Visualization

The updated knn_results.png dashboard includes two additional panels comparing KNN and Logistic Regression:

- **Accuracy, Precision, Recall & F1** — Grouped bar chart comparing all four metrics side by side for both models.
- **Confusion Matrix Counts** — Grouped bar chart comparing TP, TN, FP, FN counts directly, making the recall vs precision trade-off immediately visible.